

Course outline (Syllabus of the course)

Course Code:CS1105		
Name of the Programme: BSC (Honors)		
Name of the Course: Introduction to Version Control		
Semester: II		
Course credit	No. of hours per week	Total no. of Teaching hours
02	0:1:2	45
Course Objectives:		
a) Develop computational thinking and effective problem-solving abilities		
b) Gain hands-on experience with version control, AI tools, and modern technologies.		
c) Encourage collaborative learning and peer-to-peer interaction.		
d) Practice ethical, responsible, and reflective coding habits.		

Syllabus:

Tutorials		[15 Hours]
1	Introduction to competitive programming and coding platforms: LeetCode, HackerRank, Codeforces	
2	Benefits of practicing on coding platforms; Problem-solving strategies	
3	Introduction to Version Control Systems (VCS): Centralized vs. Distributed	
4	Installing Git on Windows/Linux/Mac; Setting up user credentials	
5	Basic Git operations: git init, git add, git commit, git status, git log	
6	Understanding local vs. remote repositories; Git hosting platforms (GitHub, GitLab, Bitbucket)	
7	Creating and linking repositories on GitHub; Pushing and pulling changes	
8	Forking and cloning repositories; Understanding gitignore	
9	Branching in Git: Creating, switching, and deleting branches	
10	Merging branches; Identifying and resolving merge conflicts	
11	Introduction to GitHub collaboration: Pull requests and code reviews	
12	Best practices in reviewing and managing pull requests	
13	Advanced Git features: git stash, git rebase vs git merge, git tag, git cherry-pick	
14	Introduction to GitHub Co-Pilot; Installing and configuring Co-Pilot in VS Code	
15	Ethical use of AI in coding; Best practices: commit messages, branching strategies, clean history	

Lab Programs		[30 Hours]
1	Create accounts on LeetCode, HackerRank, and Codeforces and solve two problems on each platform.	
2	Focus on array and string problems on LeetCode and reflect on solution strategies.	
3	Participate in a Codeforces contest and submit a solution analysis report.	

5	Install Git and configure user credentials on your system.
6	Initialize a new Git repository and commit a sample file using basic Git commands.
7	View repository status and commit history using git status and git log.
8	Create and link a local repository with GitHub and push commits using git push.
9	Pull updates from a remote repository and manage repository synchronization.
10	Fork and clone a GitHub repository, create a pull request with changes.
11	Create feature branches, switch between them, and merge them using Git.
12	Simulate and resolve a merge conflict in a collaborative project setup.
13	Use git stash to save and apply temporary changes during development.
14	Rebase a feature branch onto the main branch using git rebase
15	Tag specific commits using git tag and cherry-pick selected commits into another branch.

Course outcomes	
a)	Understand the purpose, structure, and benefits of centralized and distributed version control systems.
b)	Develop problem-solving abilities using coding platforms and real-world programming tasks.
c)	Demonstrate proficiency in Git and GitHub for collaborative version control and project management.
d)	Apply advanced Git techniques and adopt best practices to enhance workflow efficiency and team productivity.

Text books	
1.	Dan Jurafsky and James Martin. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition. Prentice Hall, Second Edition, 2009. ISBN-13: 978-0131873216.
2.	Chris Manning and Hinrich Schütze. Foundations of Statistical Natural Language Processing. MIT Press, Cambridge, MA: May 1999. ISBN-13: 978-0262133609.
3.	Steven Bird, Ewan Klein, and Edward Loper. Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit. O'Reilly Media, First Edition, June 2009. ISBN-13: 978-0596516499.

Reference books	
1.	Jacob Eisenstein. Introduction to Natural Language Processing. The MIT Press. 2019. ISBN-13: 978-0262042840.
2.	Delip Rao and Brian McMahan. Natural Language Processing with PyTorch: Build Intelligent Language Applications Using Deep Learning. O'Reilly Media. 2019. ISBN-13: 978-1491978238.

3.	Natural Language Processing - Course - Swayam – NPTEL. (https://onlinecourses.nptel.ac.in/noc19_cs56/preview)
4.	Applied Natural Language Processing- Course - Swayam – NPTEL. (https://onlinecourses.nptel.ac.in/noc20_cs87/preview)